

1. Introduction

Business Problem:

A subscription-based business is losing customers over time. The company wants to understand why customers are leaving and what factors are driving this churn, so it can take action to retain more customers.

Objective:

The objective of this analysis is to explore the customer churn dataset, identify patterns and behaviors common among customers who churn, and turn those patterns into clear business insights.

Why Churn Analysis Matters:

For a subscription business, customers pay recurring monthly or yearly fees. When customers churn, the company loses not just one payment but all future recurring revenue from that customer. It is generally far more expensive to acquire a new customer than to retain an existing one, so understanding why customers leave is directly tied to protecting business revenue and growth.

2. Dataset Overview

The dataset used for this analysis is the Telco Customer Churn dataset, which contains customer-level information for a subscription-based telecom business.

Number of Rows and Columns:

The dataset contains 7,043 rows (customers) and 21 columns (features).

Numerical Columns:

tenure, MonthlyCharges, TotalCharges

Categorical Columns:

gender, SeniorCitizen, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod

Target Column:

Churn — indicates whether the customer left the company (Yes) or stayed (No).

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Data columns (total 21 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   customerID                            7043 non-null   object
1   gender                                7043 non-null   object
2   SeniorCitizen                         7043 non-null   int64
3   Partner                                7043 non-null   object
4   Dependents                             7043 non-null   object
5   tenure                                 7043 non-null   int64
6   PhoneService                          7043 non-null   object
7   MultipleLines                         7043 non-null   object
8   InternetService                       7043 non-null   object
9   OnlineSecurity                        7043 non-null   object
10  OnlineBackup                          7043 non-null   object
11  DeviceProtection                      7043 non-null   object
12  TechSupport                           7043 non-null   object
13  StreamingTV                           7043 non-null   object
14  StreamingMovies                       7043 non-null   object
15  Contract                              7043 non-null   object
16  PaperlessBilling                      7043 non-null   object
17  PaymentMethod                         7043 non-null   object
18  MonthlyCharges                        7043 non-null   float64
19  TotalCharges                          7043 non-null   object
20  Churn                                 7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

```

3. Data Cleaning

Missing Values Found:

A formal null check (`df.isnull().sum()`) showed 0 missing values across all columns. However, the `TotalCharges` column, while technically not null, contained 11 blank space values (' ') instead of proper missing-value markers.

How Missing Values Were Handled:

The blank values in TotalCharges were replaced with NaN and the column was converted from text (object) to a proper numeric (float) type. These 11 rows were then filled with 0, since further inspection showed they all corresponded to brand-new customers with tenure = 0 who had not yet been billed.

Duplicate Rows Found:

A duplicate check (df.duplicated().sum()) returned 0 — no duplicate customer records were present in the dataset.

Data Type Corrections:

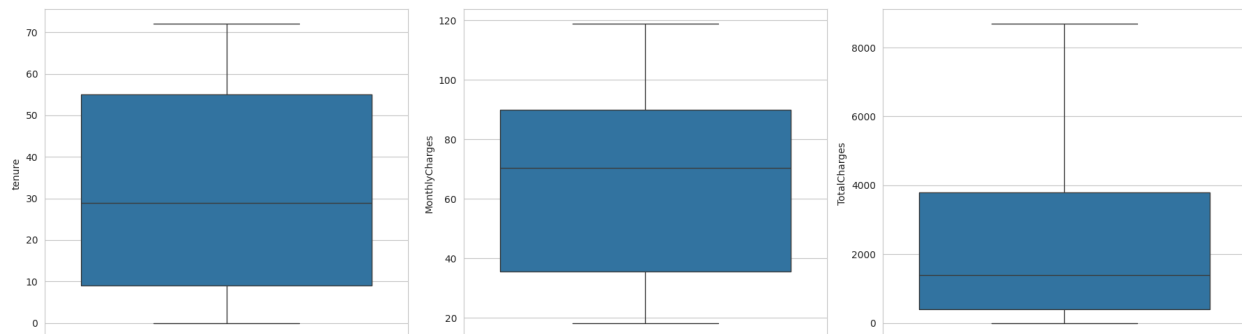
TotalCharges was converted from object (text) to float64 (numeric) so it could be used in calculations and correlation analysis. SeniorCitizen was converted from 0/1 integer values to "No"/"Yes" labels for clearer charts.

Category Cleaning:

The categorical columns (Contract, PaymentMethod, InternetService, etc.) were checked and found to already be consistently labeled, with no case mismatches or extra spaces requiring cleanup.

Outlier Observations:

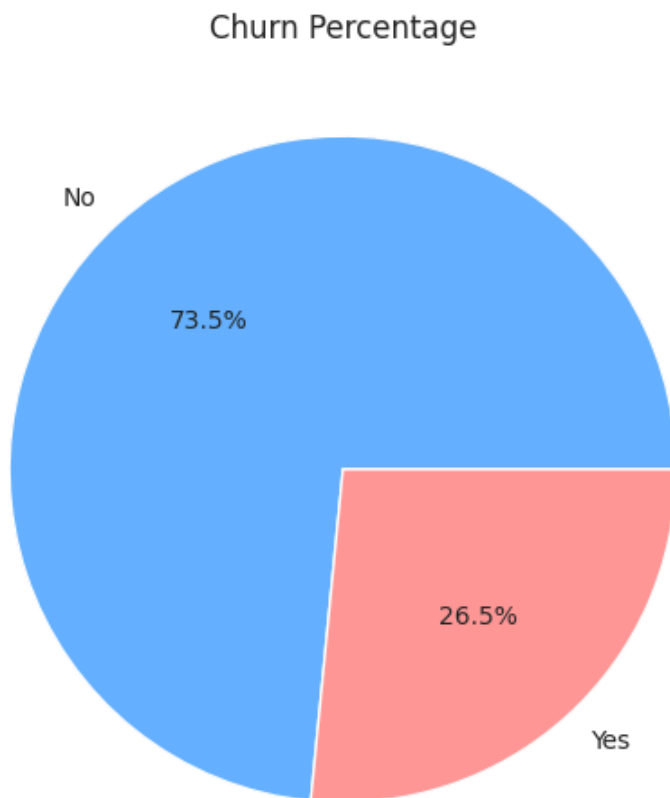
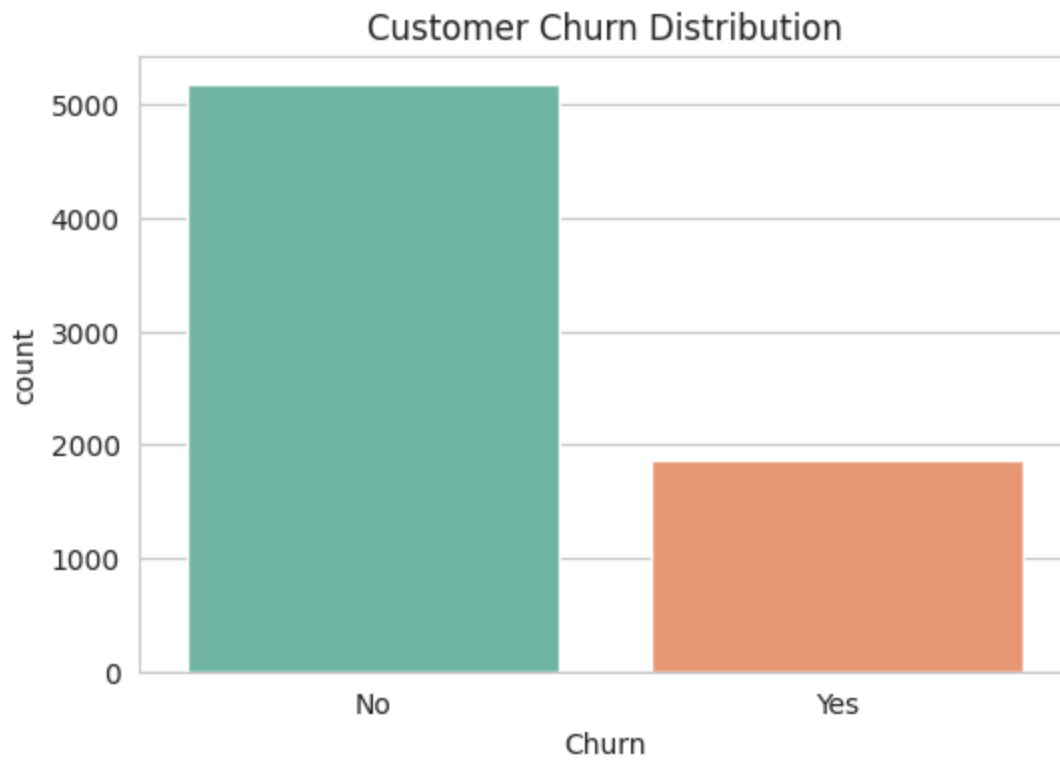
Box plots of tenure, MonthlyCharges, and TotalCharges showed no extreme outliers — values stayed within realistic ranges (tenure: 0-72 months, MonthlyCharges: \$18.25-\$118.75). This is expected since the data represents real recurring billing amounts rather than free-form entries.



4. Exploratory Data Analysis

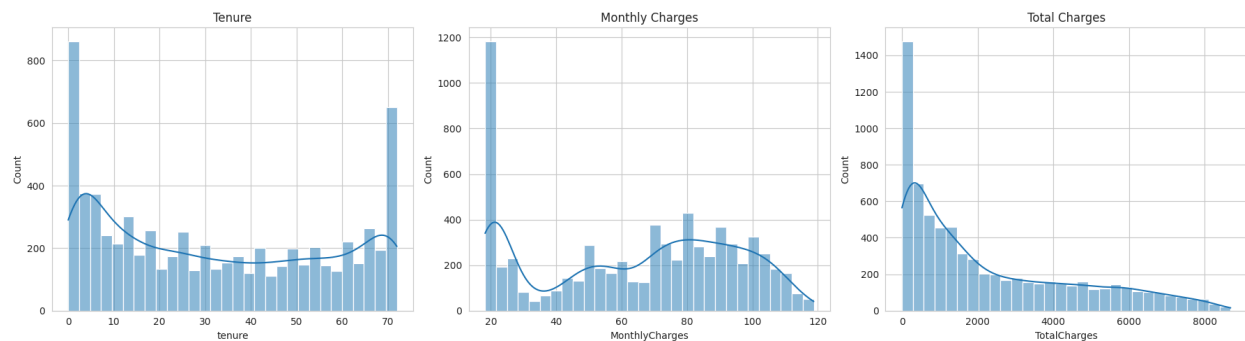
Churn Distribution:

Of the 7,043 customers, 5,174 (73.46%) have stayed and 1,869 (26.54%) have churned. This is a moderately imbalanced target, but with enough churned customers to identify meaningful patterns.



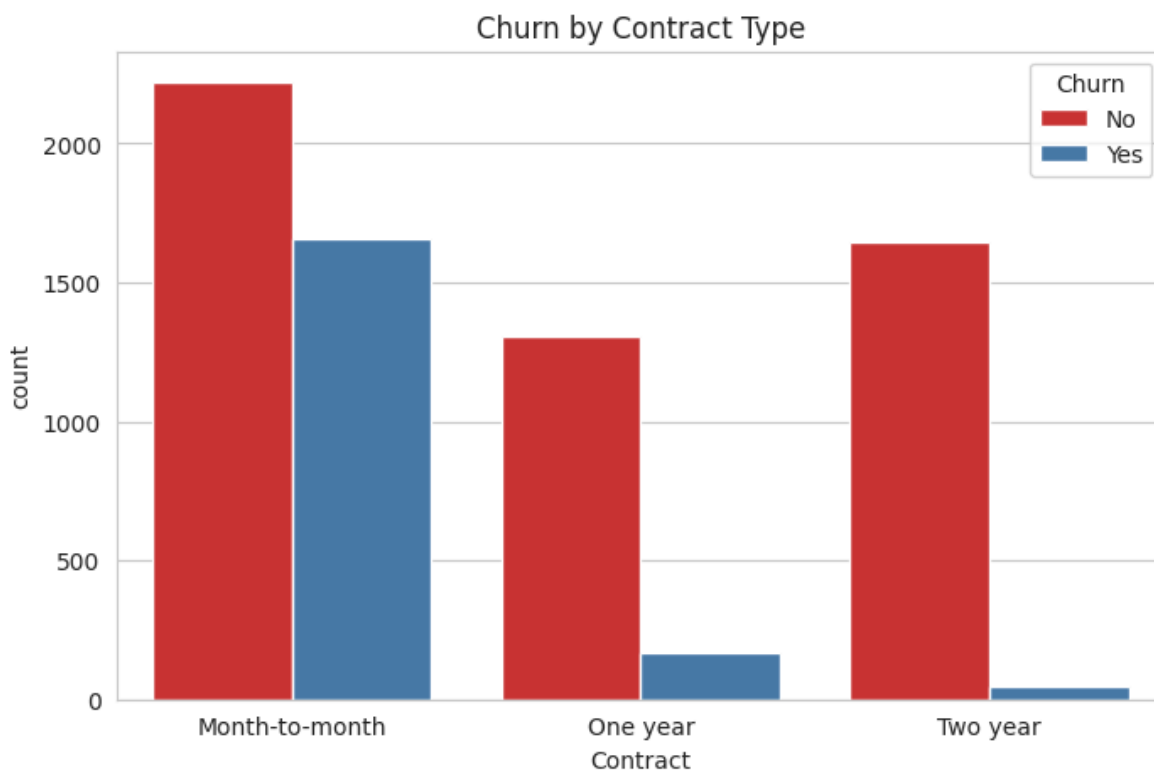
Tenure Analysis:

The tenure distribution is roughly U-shaped — a large number of customers have very low tenure (just joined) and another large group has tenure close to 72 months (long-term, loyal customers), with fewer customers in between. The skew value of 0.24 confirms the distribution is fairly balanced rather than heavily skewed in one direction.



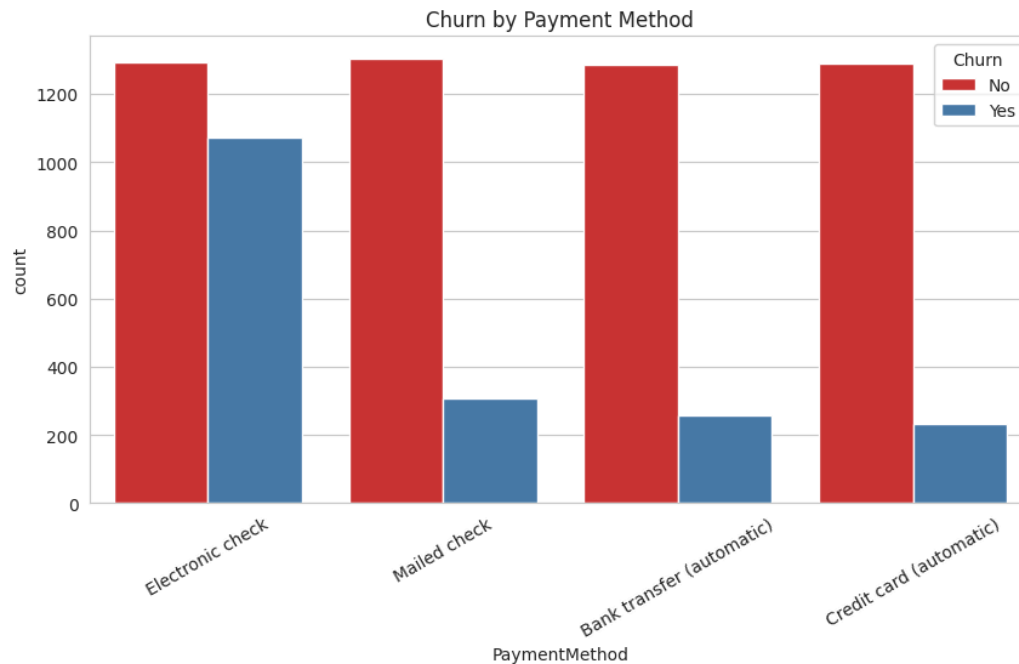
Contract Type Analysis:

Customers on month-to-month contracts show a much higher proportion of churn compared to customers on one-year or two-year contracts. Two-year contract customers show the lowest churn by far.



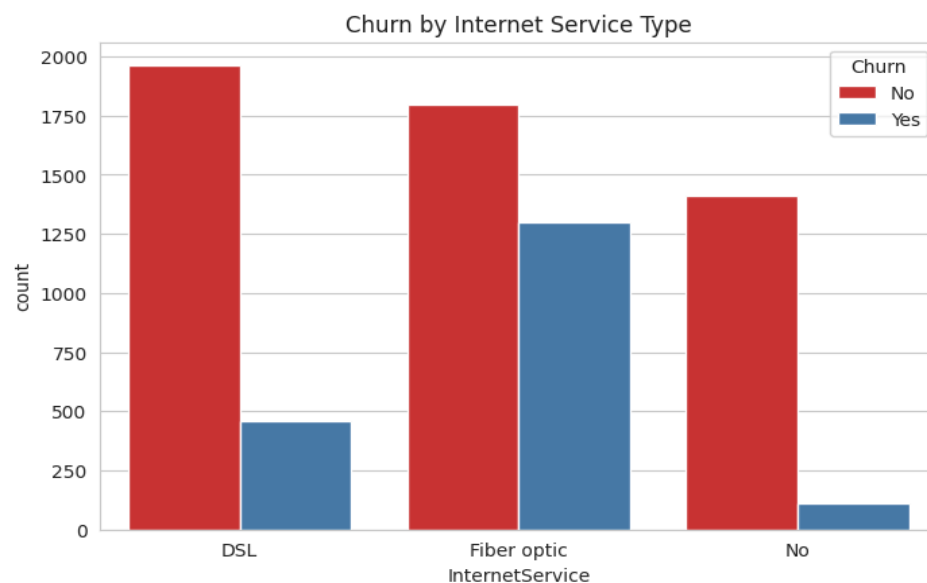
Payment Method Analysis:

Customers paying via Electronic check show a noticeably higher proportion of churn compared to the other three payment methods (Mailed check, Bank transfer automatic, Credit card automatic), which all show similarly low churn proportions.



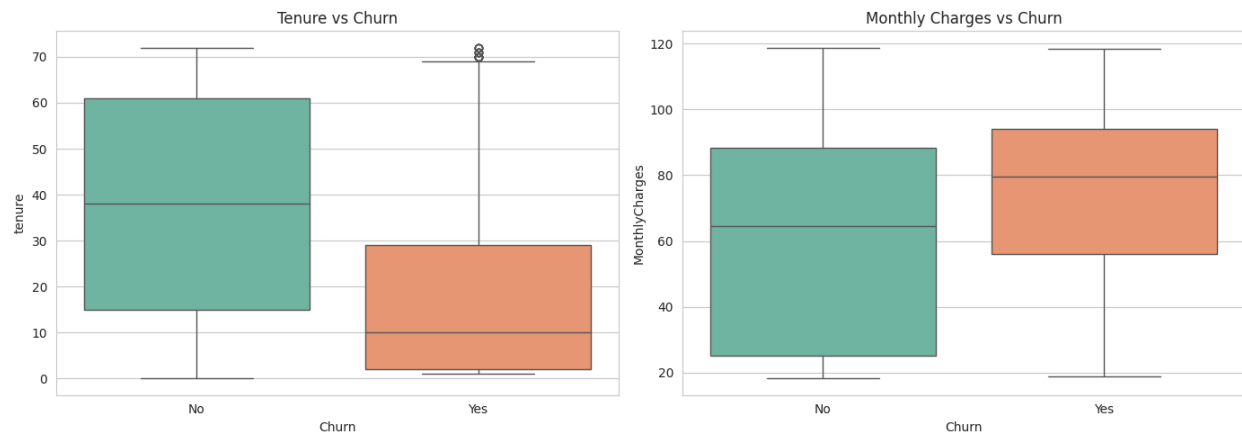
Internet Service Analysis:

Customers with Fiber optic internet show a much higher churn proportion than DSL customers, and customers with No internet service show the lowest churn of all three groups.



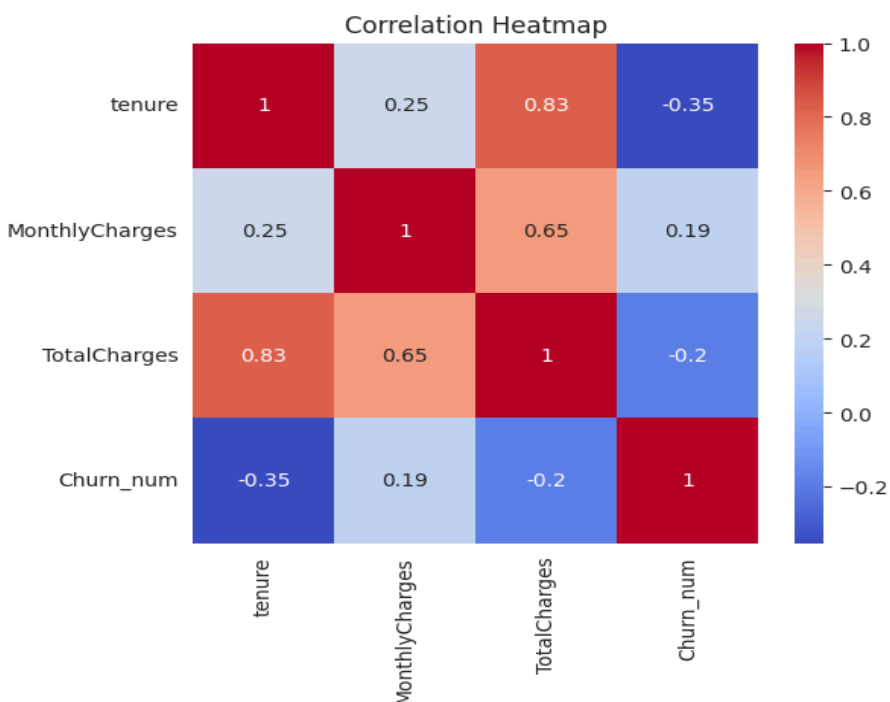
Monthly Charges and Tenure vs Churn:

Comparing tenure and monthly charges directly against churn status shows that churned customers tend to have lower tenure and somewhat higher monthly charges than customers who stayed.



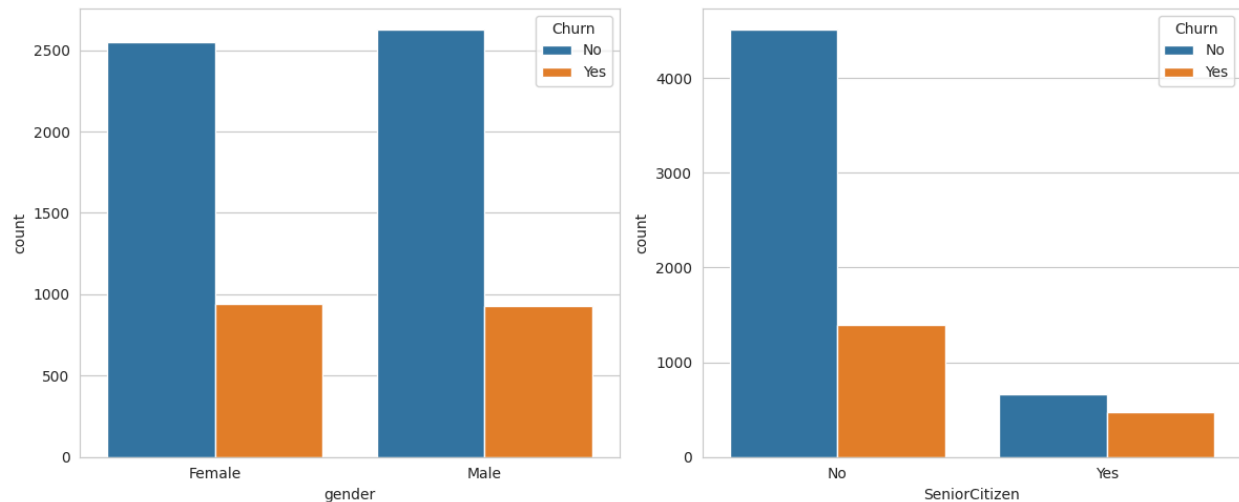
Correlation Analysis:

A correlation heatmap was built using tenure, MonthlyCharges, TotalCharges, and a numeric version of Churn. The results show tenure has a -0.35 correlation with churn (longer-tenured customers churn less), MonthlyCharges has a +0.19 correlation with churn (higher-paying customers churn slightly more), and TotalCharges has a -0.2 correlation with churn. Tenure and TotalCharges are strongly correlated with each other (0.83), which makes sense since total charges accumulate the longer a customer stays.



Demographics:

Gender shows almost no difference in churn rate between male and female customers. Senior citizens show a proportionally higher churn rate compared to non-senior customers.



5. Key Business Insights

Insight 1: Month-to-Month Contracts Drive Higher Churn

Observation: Customers on month-to-month contracts churn at a much higher rate than customers on one-year or two-year contracts.

Possible Reason: Month-to-month customers have no long-term commitment and can cancel at any time without penalty.

Business Impact: The company risks losing a large share of revenue from short-term customers who can leave with minimal friction.

Recommendation: Offer discounts or added benefits to encourage customers to switch to longer-term contracts.

Insight 2: Electronic Check Users Churn More

Observation: Customers paying via Electronic check churn more than those using automatic payment methods.

Possible Reason: Manual payment methods may create more friction (forgotten payments, more conscious monthly decision to continue) compared to automatic, "set and forget" payments.

Business Impact: Customers on manual payment methods are a higher churn-risk segment.

Recommendation: Encourage customers to switch to automatic payment methods through incentives or simplified sign-up flows.

Insight 3: Fiber Optic Customers Churn More Despite Being a Premium Service

Observation: Fiber optic internet customers churn more than DSL customers.

Possible Reason: Fiber optic is typically priced higher, and may also carry higher customer expectations around speed and reliability that aren't always met.

Business Impact: A premium service line is losing customers at a higher rate than the more basic offering, which is a concerning trend for revenue from the higher-margin product.

Recommendation: Investigate fiber service quality and pricing, and consider satisfaction surveys targeted at fiber customers.

Insight 4: Low-Tenure Customers Are at Higher Risk

Observation: Churned customers tend to have lower tenure than retained customers.

Possible Reason: New customers haven't yet built loyalty or fully experienced the value of the service.

Business Impact: The first few months of the customer relationship are the highest-risk period for losing that customer permanently.

Recommendation: Strengthen onboarding with welcome guidance, early check-ins, and proactive support during the first 3-6 months.

Insight 5: Higher Monthly Charges Are Linked to Higher Churn

Observation: Churned customers tend to have somewhat higher monthly charges, and MonthlyCharges shows a positive correlation with churn.

Possible Reason: Price-sensitive customers may decide the service isn't worth the cost, especially early in their tenure before they've seen long-term value.

Business Impact: Higher-paying customers, who are valuable to retain, are at elevated risk of leaving.

Recommendation: Offer loyalty discounts or flexible plan downgrades instead of losing higher-paying customers outright.

6. Feature Engineering Ideas

1. TenureGroup — Bucket tenure into "New" (0-12 months), "Medium" (13-36 months), and "Long-term" (37+ months). Helps models capture the non-linear relationship between tenure and churn risk seen in this analysis.

2. HighMonthlyCharge — A flag marking customers above the median monthly charge. Useful since higher charges showed a positive correlation with churn.

3. AutoPaymentFlag — A flag for whether the customer uses an automatic payment method (bank transfer or credit card) versus manual (electronic check or mailed check). Useful since manual payment methods showed higher churn.

4. ContractRiskScore — A simple numeric encoding of contract type risk

(month-to-month = high risk, one year = medium, two year = low), since contract type was one of the strongest churn signals found.

5. FiberCustomerFlag — A flag for customers on fiber optic internet, since this group showed disproportionately higher churn than DSL or no-internet customers.

7. Conclusion

This exploratory analysis of 7,043 telecom customers found that churn is concentrated among a few clear customer segments: those on month-to-month contracts, those paying via electronic check, those using fiber optic internet, and those with low tenure. Monthly charges also show a positive relationship with churn, suggesting price sensitivity plays a role.

The most important risk factors identified are contract type and tenure, both of which show strong, intuitive relationships with churn (shorter commitment and shorter relationship length both increase churn risk).

These findings provide a strong foundation for future machine learning modeling. Features like contract type, payment method, internet service type, tenure, and monthly charges all show clear relationships with churn and would likely be valuable predictors in a churn prediction model.